THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 8: **Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of the first 25 odd natural numbers is: 1

(a) 600 **(b)** 675 **(c)** 650 **(d)** 625

2. The ratio of the 5th term to the 3rd term of a GP is 25. The common ratio (positive) is: 1

(a) 7 **(b)** 5 **(c)** 6 **(d)** 4

3. The number of multiples of 6 between 41 and 300 (both excluded) is: 1

(a) 45 **(b)** 44 **(c)** 42 **(d)** 43

4. The value of the 20th term minus the 10th term of the AP $12, 17, 22, 27, \ldots$ is: 1

(a) 55 **(b)** 45 **(c)** 60 **(d)** 50

5. In an AP, $t_1 = -10$ and $t_{14} = 81$. The common difference is: 1

(a) 9 **(b)** 7 **(c)** 6 **(d)** 8

6. In an AP, $d = 6$ and $t_{15} = 76$. The first term is: 1

(a) -6 **(b)** -9 **(c)** -8 **(d)** -7

7. The 4th and 9th terms of an AP are 5 and 20. Its 15th term is: 1

(a) 37 **(b)** 39 **(c)** 40 **(d)** 38

8. The sum of the first 35 even natural numbers is: 1

(a) 1225 **(b)** 1260 **(c)** 1330 **(d)** 1295

9. The sum of the first 32 odd natural numbers is: 1

(a) 992 **(b)** 1088 **(c)** 1024 **(d)** 1056

10. The ratio of the 5th term to the 3rd term of a GP is 36. The common ratio (positive) is: 1

(a) 7 **(b)** 8 **(c)** 5 **(d)** 6

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The second and fifth terms of a GP are 12 and 768. Find the common ratio and the first term. 2

12. The 5th term of an AP is 7 and the 12th term is 35. Find d. 2

13. Find the sum of all multiples of 7 between 40 and 180. 2

14. The second and fifth terms of a GP are 5 and 625. Find the common ratio and the first term. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the number of terms of the AP $4, 15, 26, \ldots$ whose last term is 191, and its sum. 3

16. The 6th term of an AP is 37 and the 13th term is 58. Find the AP and its 30th term. 3

17. Find the sum of the first 10 terms of the AP whose first term is 20 and whose 5th term is 40. 3

OR

Find the number of terms of the AP $60, 56, 52, \ldots$ whose last term is -44, and its sum. 3

18. The 5th term of an AP is 26 and the 12th term is -16. Find the AP and its 22nd term. 3

OR

Find the sum of the first 18 terms of the AP whose first term is 7 and whose 11th term is 67. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A ladder has 21 rungs. The bottom rung is 80 cm long and each rung above is 2 cm shorter than the one below. Find the length of the top rung and the total length of wood needed for the rungs. 5

OR

A sum of ₹40000 is invested at 10% per year compound interest. Show that the amounts at the end of years form a GP and find the amount after 2 years. 5

20. A ladder has 17 rungs. The bottom rung is 200 cm long and each rung above is 8 cm shorter than the one below. Find the length of the top rung and the total length of wood needed for the rungs. 5

OR

A sum of ₹25000 is invested at 20% per year compound interest. Show that the amounts at the end of years form a GP and find the amount after 2 years. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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